

# QuadRAG: A Four-Index Multimodal Retrieval-Augmented Framework for Video Understanding

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*Abstract*—This abstract is a temporary placeholder for the preliminary report. The final version will briefly introduce the motivation behind the project, outline the proposed quad-modal retrieval approach, and summarize the key experimental findings from the Video-MME, along with the overall conclusions.

## I. INTRODUCTION

Large Language Models (LLMs) have achieved significant success in a short period of time, yet their capabilities don't fully fill the gap created by rapidly increasing video content. Several studies have been conducted to use LLMs for comprehending and processing videos [1], [2], [3], [4], [5]; these are collectively termed Large Video-Language Models (LVLMs). LVLMs have shown promising results for short videos, but their ability to understand and comprehend long videos remains a major problem.

To address this challenge, fine-tuning LVLMs has been used in many studies [6], [7], [8] to understand long videos. This approach has significant drawbacks: fine-tuning LVLMs is highly resource-intensive, requiring large, high-quality datasets and substantial computational power. Retrieval-Augmented Generation (RAG) [10] is a technique of using relevant documents from an external corpus to enhance generative tasks, thus improving response quality in LLMs. The implementation of RAG in video-based tasks has been used in recent studies [11], [12], fetching context from videos and sending it to a generative model; this is known as the GPT-based Agent method. It is a powerful method, yet it does not produce the niche-specific results accurately.

To address these limitations, we need more efficient, cost-effective, and adaptive solutions. RAG has emerged as a powerful, training-free alternative [9]. In this project, we are developing an advanced video comprehension system by building on and extending the Video-RAG concept.

We introduce innovations in the RAG architecture itself. While prior work [9] focuses on converting all video information (including object detection and optical character recognition (OCR)) into "visually-aligned auxiliary texts," our project uses a novel pipeline that creates and leverages four distinct, parallel semantic indexes:

- 1) A **Image Index** from raw video frames.
- 2) An **Audio Index** from transcribed spoken dialogue.

- 3) A **Description Index** from raw video frames.
- 4) A **Domain Captions Index** from raw video frames and context engineering techniques.

Our hypothesis is that this quad-modal, modality-specific pipeline will provide a richer and more comprehensive context to the LLM generator. This allows our system to answer more accurately by capturing information from the most relevant source, whether it be a visual detail, a spoken phrase, or a written caption. This system allows user to extract more granular, niche-specific information from the videos with the help of domain specific embeddings.

## II. RELATED WORK

Our project is situated at the intersection of video-language modeling and information retrieval. It is inspired by the Video-RAG paper [9] and implemented with RAG architectural changes. The related work, as categorized by Luo [9], is as follows:

### A. Large Video-Language Models (LVLMs)

LVLMs, such as Video-ChatGPT [13], VideoChat [14], and Video-LLaVA [15], have shown promising results and performance by aligning pre-trained visual encoders with LLMs. However, they face the challenge of a limited number of video frames they can process, limiting their effectiveness on long-form videos.

### B. Long-context Large Video-Language Models

To address the limitations of standard LVLMs, models like LongVA [16] and Long-LLaVA [17] use the approach of extending the context window to process more frames. This technique faces challenges with high computational costs and information redundancy. Performance can even degrade when sampling too many frames, suggesting a bottleneck in this strategy [9].

### C. GPT-based Agent Video Understanding

Agentic frameworks, such as VideoAgent [18], DrVideo [19], and OmAgent [20], use a powerful LLM (e.g., GPT-4) to orchestrate external tools for video analysis. While highly capable, this method is resource-intensive, slow, and heavily dependent on closed-source APIs, which are significantly costly and also limit research flexibility [9].

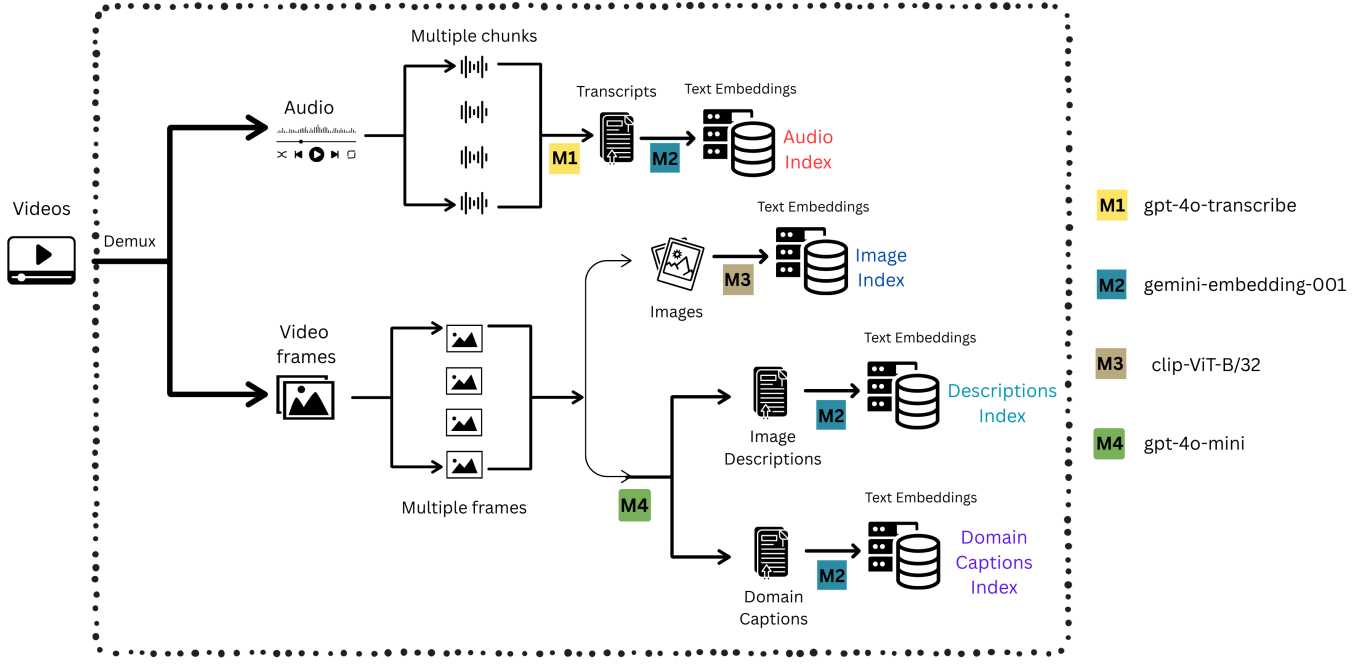


Fig. 1: Proposed Quad-Modal RAG Architecture. This diagram shows how video, audio, and text streams are independently indexed and later retrieved to generate an answer.

#### D. Retrieval-Augmented Generation for Video

RAG-based pipelines have been proposed as a cost-effective and training-free alternative. The "Video-RAG" paper [9] is a state-of-the-art example. Its method involves the utilization of external tools to generate text-based representations of audio (ASR), on-screen text (OCR), and detected objects (DET). These "auxiliary texts" are then retrieved to augment an LVLm. Our work is inspired by this but differs fundamentally in our retrieval architecture, as detailed in the following section.

### III. METHOD

Our proposal is a "training-free" systems integration project. We are not training any models from scratch. Instead, we are building a pipeline that integrates state-of-the-art, pre-trained open-source models to create a new video comprehension capability.

Our proposed architecture, illustrated in Figure 1, is designed to ingest, process, and index video content across three distinct modalities.

The ingestion and retrieval pipeline consists of the following steps:

- 1) **Preprocessing & Modality Separation:** A given video file is loaded. The audio stream is extracted, and the corresponding subtitle/caption file is parsed.
- 2) **Visual Stream Indexing:** The video stream is sampled into frames. Each frame is passed through a pre-trained visual-language model (e.g., CLIP) to generate vector embeddings that capture its visual content.

- 3) **Audio Stream Indexing:** The extracted audio stream is transcribed into text using a pre-trained speech-recognition model (gpt-4o-transcribe). The resulting transcript is segmented and embedded using an embedding model like gemini-embedding-001.
- 4) **Description Indexing:** Frames are sampled and with the help of vision models like gpt-4o-mini, the descriptions are generated and then embeddings are created and stored using models like gemini-embedding-001.
- 5) **Domain Captions Indexing:** The same sampled images are taken and with the help of vision models like gpt-4o-mini, and using some context engineering the descriptions are generated and then embeddings are created and stored using models like gemini-embedding-001.
- 6) **Unified Vector Store:** All four sets of embeddings are stored in a single vector database (e.g., ChromaDB, PixelTable), with metadata attached to each vector identifying its source modality ('visual', 'audio', 'descriptions', 'domain caption') and its timestamp.
- 7) **Multimodal RAG and Generation:** When a user submits a query, the video is embedded with the audio, image, and description indexes. The agent then asks the user to provide the domain specific instructions. To which user provides a description like "Capture the emotions" (emotion intelligence problem; see Figure 2) or "understand where the objects are placed" (object detection problem) or "identify when you see the candidates look away from the computer" (Examination fraud identification problem). This information is taken and dynamically the Domain captions embeddings are

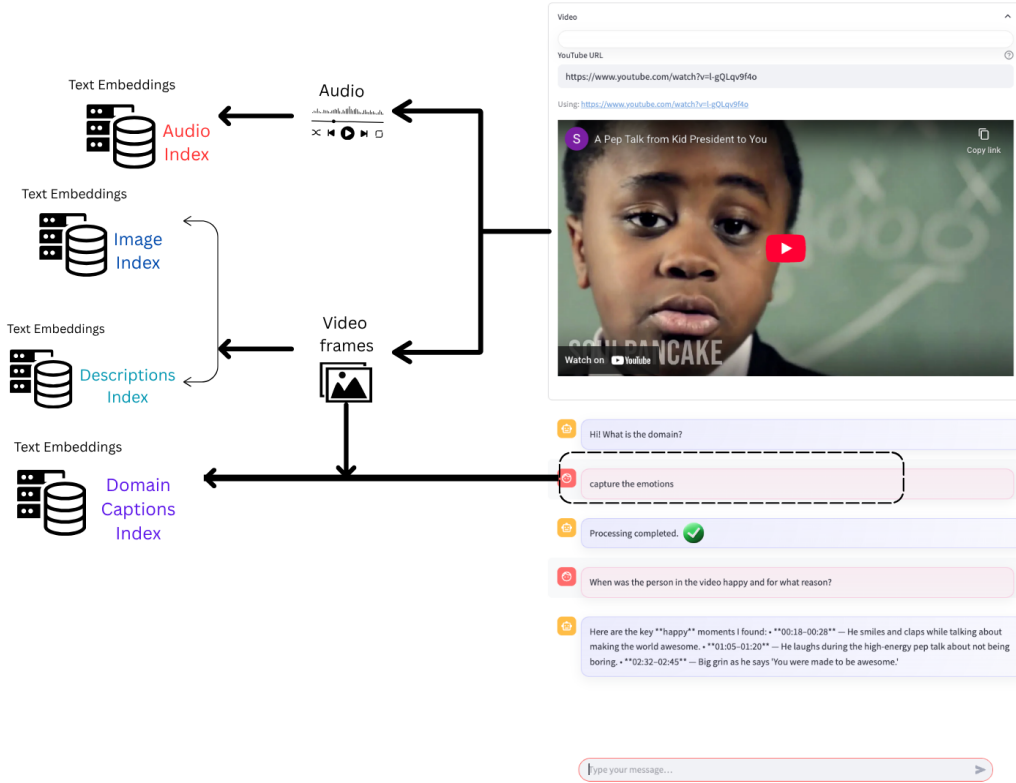


Fig. 2: Example Chat in Streamlit along with the background RAG process

created. Then the user query is embedded and then used to search against all four indexes simultaneously. The top-K results from each modality are retrieved, "fused" to create a rich context, and fed into a generator LLM to produce the final answer. As you can see this method solves multiple identification problems specific to user interested domains.

#### IV. EXPERIMENTS AND RESULTS

##### A. Datasets

To ensure a clear and direct comparison with prior state-of-the-art work, we will evaluate our pipeline on the same public academic benchmarks used in the Video-RAG paper [9]. This includes Video-MME [21], a well-known benchmark for evaluating the performance of LVLMS. This dataset is divided into three subsets based on video length (ranging from 11 seconds to 1 hour).

1) *Base Dataset: Video-MME*: We are using the Video-MME benchmark for our experiments. It is a large-scale evaluation dataset designed for multi-modal long-video understanding. Video-MME consists of 900 publicly available YouTube videos spanning six domains and 30 different sub-categories, annotated with 2700 multiple-choice question-answer pairs (three per video). The dataset is divided into 3 duration categories—short ( $\leq 30$  seconds), medium (1-10 mins), and

long ( $\geq 10$  mins)—to enable evaluation across different time durations. The official metadata, including YouTube identifiers, video durations, question text, answer options, and correct labels, is distributed through the project's Hugging Face repository (Imms-lab/Video-MME).

2) *Construction of 10% Data-Subset*: To reduce computational cost while preserving class diversity, we derived a 10% subset of Video-MME. We selected 30 short, 30 medium, and 30 long videos using random sampling with a fixed seed to ensure reproducibility. Each video retains its original three question-answer pairs, resulting in 90 videos (around 25 hours of content) and 270 QA pairs in total. This stratified sampling maintains the original distribution of domains and difficulty, ensuring a fair comparison against published baseline works.

3) *Metadata Extraction and Organization*: We extracted the official Video-MME metadata using the `datasets` library, grouped entries by unique video identifier, and aggregated all associated QA annotations per video. The resulting manifests comprise:

- 1) `videomme_subset_videos.csv` – list of selected video IDs, durations, and source URLs;
- 2) `videomme_subset_qa.jsonl` – structured question-answer annotations (270 records);
- 3) `urls.txt` – plain text list of YouTube links for video retrieval.

All metadata files are lightweight ( $< 2$  MB) and distributed as part of our supplementary materials to facilitate exact replication.

4) *Video Retrieval and Storage*: The YouTube videos were retrieved at  $\leq 480p$  resolution to balance fidelity and storage efficiency. To comply with Video-MME’s research-only license, we accessed videos solely through their public URLs and did not redistribute any media content. Each downloaded video has its own metadata and subtitle files.

5) *Reproducibility*: To ensure our work is reproducible, all sampling and retrieval procedures are fully deterministic. The manifest files, scripts, and notebook used to generate this 10% subset are available in our supplementary repository. This allows future studies to reconstruct the identical subset and validate our results.

6) *Download Results*: During our data retrieval process, we successfully downloaded 89 of the 90 targeted videos; one video was no longer available. The downloaded package for each video included several key files, as summarized in Table I.

| File / Status  | Explanation                                                                |
|----------------|----------------------------------------------------------------------------|
| 89 videos      | One video became unavailable or restricted. The rest downloaded correctly. |
| .vtt files     | Subtitle transcripts (uploader-provided or auto-generated).                |
| .info.json     | Per-video metadata (title, description, etc.).                             |
| downloaded.txt | Download ledger to resume or skip completed items.                         |

TABLE I: Summary of Video-MME 10% subset retrieval.

### B. Implementation Details

(This section is a placeholder. The final report will detail the specific models, libraries (e.g., PyTube, Gemini and OpenAI models, Other open source models, Sentence-Transformers, ChromaDB, PixelTable), and any other hardware used for the implementation.)

### C. Evaluation Metrics

(This section is a placeholder. As per the guidelines, we will use standard metrics to evaluate our retrieval system, as shown in Table II.)

| Metric         | Description                                                                                                                               |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Top-K Accuracy | The percentage of queries where the correct contextual segment (video, audio, or text) is ranked within the top K results (e.g., $K=3$ ). |
| MRR            | Mean Reciprocal Rank. A metric that scores how high the first correct retrieved segment is ranked, averaging across all queries.          |

TABLE II: Planned Quantitative Evaluation Metrics.

### D. Results

(This section is a placeholder for the final report. It will contain the quantitative results from our experiments, comparing our multi-modal RAG pipeline against baselines on the Video-MME 10% subset.)

## V. CONCLUSIONS

(This section is a placeholder for the final report. The conclusion will summarize what we learned/concluded from the project, analyze the effectiveness of our multi-modal approach versus the auxiliary-text method, and discuss limitations and potential future work.)

## VI. PEER EVALUATION

(This section is a placeholder. As per the final report guidelines, we will include a table outlining the contribution of each team member.)

| Team Member            | Contribution Score (/20) |
|------------------------|--------------------------|
| Kamil Akhtar           | (TBD)                    |
| Sanjay Reddy Choudapur | (TBD)                    |

TABLE III: Self-Peer Evaluation Table.

## REFERENCES

- [1] Lin Chen, Xilin Wei, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Bin Lin, Zhenyu Tang, et al. Sharegpt4video: Improving video understanding and generation with better captions. *arXiv preprint arXiv:2406.04325*, 2024. 1
- [2] Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, et al. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24185–24198, 2024. 1
- [3] KunChang Li, Yanan He, Yi Wang, Yizhuo Li, Wenhai Wang, Ping Luo, Yali Wang, Limin Wang, and Yu Qiao. Videochat: Chat-centric video understanding. *arXiv preprint arXiv:2305.06355*, 2023. 1
- [4] Ji Lin, Hongxu Yin, Wei Ping, Pavlo Molchanov, Mohammad Shoeybi, and Song Han. Vila: On pre-training for visual language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 26689–26699, 2024. 1
- [5] Jiajun Liu, Yibing Wang, Hanghang Ma, Xiaoping Wu, Xiaoqi Ma, xiaoming Wei, Jianbin Jiao, Enhua Wu, and Jie Hu. Kangaroo: A powerful video-language model supporting long-context video input. *arXiv preprint arXiv:2408.15542*, 2024. 1
- [6] Yuzhang Shang, Bingxin Xu, Weitai Kang, Mu Cai, Yuheng Li, Zehao Wen, Zhen Dong, Kurt Keutzer, Yong Jae Lee, and Yan Yan. Interpolating video-llms: Toward longer sequence llms in a training-free manner. *arXiv preprint arXiv:2409.12963*, 2024. 1
- [7] Xidong Wang, Dingjie Song, Shunian Chen, Chen Zhang, and Benyou Wang. Longlava: Scaling multi-modal llms to 1000 images efficiently via hybrid architecture. *arXiv preprint arXiv:2409.02889*, 2024. 1
- [8] Yongshuo Zong, Ismail Elezi, Yongxin Yang, Jiankang Deng, and Timothy Hospedales. Long-context vision large language models: Empirical insights and a baseline. In *Workshop on Long Context Foundation Models*, 2024. 1
- [9] Y. Luo, X. Zheng, X. Yang, et al., "Video-RAG: Visually-aligned Retrieval-Augmented Long Video Comprehension," *arXiv:2411.13093v3 [cs.CV]*, 20 Dec 2024. 1, 2, 3
- [10] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474, 2020. 1
- [11] Md Adnan Arefeen, Biplob Debnath, Md Yusuf Sarwar Uddin, and Srimat Chakradhar. irag: An incremental retrieval augmented generation system for videos. *arXiv preprint arXiv:2404.12309*, 2024. 1
- [12] Ziyu Ma, Chenhui Gou, Hengcan Shi, Bin Sun, Shutao Li, Hamid RezaTofighi, and Jianfei Cai. Dvideo: Document retrieval based long video understanding. *arXiv preprint arXiv:2406.12846*, 2024. 1
- [13] M. Maaz, H. Rasheed, S. Khan, and F. S. Khan, "Video-chatgpt: Towards detailed video understanding via large vision and language models," *arXiv preprint arXiv:2306.05424*, 2023. 1
- [14] K. Li, Y. He, Y. Wang, et al., "Videochat: Chat-centric video understanding," *arXiv preprint arXiv:2305.06355*, 2023. 1

- [15] B. Lin, B. Zhu, Y. Ye, et al., "Video-llava: Learning united visual representation by alignment before projection," *arXiv preprint arXiv:2311.10122*, 2023. 1
- [16] P. Zhang, K. Zhang, B. Li, et al., "Long context transfer from language to vision," *arXiv preprint arXiv:2406.16852*, 2024. 1
- [17] Y. Song, W. Chen, and E. Duthie, "aws-prototyping/long-llava-qwen2-7b," 2024. 1
- [18] Y. Fan, X. Ma, R. Wu, et al., "Videoagent: A memory-augmented multimodal agent for video understanding," in *European Conference on Computer Vision*, 2025. 1
- [19] Z. Ma, C. Gou, H. Shi, et al., "Drvideo: Document retrieval based long video understanding," *arXiv preprint arXiv:2406.12846*, 2024. 1
- [20] L. Zhang, T. Zhao, H. Ying, et al., "Omagent: A multi-modal agent framework for complex video understanding with task divide-and-conquer," *arXiv preprint arXiv:2406.16620*, 2024. 1
- [21] Chaoyou Fu, Yuhan Dai, Yondong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. *arXiv preprint arXiv:2405.21075*, 2024. 3